KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5201 | 5/6 Credits: 4 | Program Core/ Elective | Source-grounded

Signals and Systems

 To introduce students to the idea of signals and systems, their characteristics in time

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5201 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5201.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Electronics and Communication/ Biomedical Engineering
Course code	5201
Course title	Signals and Systems
Semester	5/6 Credits: 4
Course category	Program Core/ Elective
Periods per week	4 (L:3, T:1, P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To introduce students to the idea of signals and systems, their characteristics in time
- and frequency domain.
- To provide basic knowledge on Fourier representation and Laplace transform and its applications on signals and systems

Course outcomes

CO	Official outcome	Study level
CO1	14 Understanding mathematical properties of signals Classify and compare continuous time and discrete	See official syllabus
CO2	12 Understanding time systems	See official syllabus
CO3	Explain Fourier representation of signals 16 Understanding Apply Laplace transform to demonstrate the	See official syllabus
CO4	16 Applying concepts of signals and systems	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	of signals	4	As specified
Module 2	Classify and compare continuous time and discrete time systems	4	As specified
Module 3	Explain Fourier representation of signals	4	As specified
Module 4	systems	4	As specified

MODULE 1

# of signals

This module is presented from the official syllabus scope for Signals and Systems. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: of signals
- Official duration: As specified
- Official topic entries captured: 4

- The full source excerpt is preserved below for coverage checking.

Define signals and its importance. 3 Understanding Explain the basic elementary signals used in

Define signals and its importance. 3 Understanding Explain the basic elementary signals used in is an official topic in Module 1 of Signals and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define signals and its importance. 3 Understanding Explain the basic elementary signals used in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define signals and its importance. 3 understanding explain the basic elementary signals used in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define signals and its importance. 3 Understanding Explain the basic elementary signals used in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Define signals and its importance. 3 Understanding Explain the basic elementary signals used in □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

4 Understanding communication.

4 Understanding communication. is an official topic in Module 1 of Signals and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding communication. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding communication. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding communication. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 4 Understanding communication. □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Select various types of signals. 4 Apply

Select various types of signals. 4 Apply is an official topic in Module 1 of Signals and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Select various types of signals. 4 Apply using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, select various types of signals. 4 apply should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Select various types of signals. 4 Apply means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Select various types of signals. 4 Apply [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain the basic operation on signals. 3 Understanding Define signals: state the importance of signals and systems in the field of science and engineering. Basic elementary signals: Unit step function, unit impulse function, ramp, parabolic, signum, exponential, rectangular, triangular, and sinusoidal. Classification of signals: continuous time and discrete time, deterministic and non- deterministic, even and odd, periodic and aperiodic, energy and power, real and imaginary. Mathematical operations on signals: amplitude scaling, time scaling, time shifting, time reversal, addition, subtraction, multiplication and convolution.

Explain the basic operation on signals. 3 Understanding Define signals: state the importance of signals and systems in the field of science and engineering. Basic elementary signals: Unit step function, unit impulse function, ramp, parabolic, signum, exponential, rectangular, triangular, and sinusoidal. Classification of signals: continuous time and discrete time, deterministic and non- deterministic, even and odd, periodic and aperiodic, energy and power, real and imaginary. Mathematical operations on signals: amplitude scaling, time scaling, time shifting, time reversal, addition, subtraction, multiplication and convolution. is an official topic in Module 1 of Signals and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the basic operation on signals. 3 Understanding Define signals: state the importance of signals and systems in the field of science and engineering. Basic elementary signals: Unit step function, unit impulse function, ramp, parabolic, signum, exponential, rectangular, triangular, and sinusoidal. Classification of signals: continuous time and discrete time, deterministic and non- deterministic, even and odd, periodic and aperiodic, energy and power, real and imaginary. Mathematical operations on signals: amplitude scaling, time scaling, time shifting, time reversal, addition, subtraction, multiplication and convolution. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the basic operation on signals. 3 understanding □ define signals: state the importance of signals and systems in the field of science and engineering. □ basic elementary signals: unit step function, unit impulse function, ramp, parabolic, signum, exponential, rectangular, triangular, and sinusoidal. □ classification of signals: continuous time and discrete time, deterministic and non- deterministic, even and odd, periodic and aperiodic, energy and power, real and imaginary. □ mathematical operations on signals: amplitude scaling, time scaling, time shifting, time reversal, addition, subtraction, multiplication and convolution. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the basic operation on signals. 3 Understanding □ Define signals: state the importance of signals and systems in the field of science and engineering. □ Basic elementary signals: Unit step function, unit impulse function, ramp, parabolic, signum, exponential, rectangular, triangular, and sinusoidal. □ Classification of signals: continuous time and discrete time, deterministic and non- deterministic, even and odd, periodic and aperiodic, energy and power, real and imaginary. □ Mathematical operations on signals: amplitude scaling, time scaling, time shifting, time reversal, addition, subtraction, multiplication and convolution. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Explain the basic operation on signals. 3 Understanding □ Define signals: state the importance of signals and systems in the field of science and engineering. □ Basic elementary signals: Unit step function, unit impulse function, ramp, parabolic, signum, exponential, rectangular, triangular, and sinusoidal. □ Classification of signals: continuous time and discrete time, deterministic and non- deterministic, even and odd, periodic and aperiodic, energy and power, real and imaginary. □ Mathematical operations on signals: amplitude scaling, time scaling, time shifting, time reversal, addition, subtraction, multiplication and convolution. ▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯ definition/meaning ▯▯▯▯▯▯▯▯▯▯▯▯▯▯. ▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯, steps, application ▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯. Technical terms English-▯ ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

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of signals

  M1.01      Define signals and its importance.                          3
Understanding

             Explain the basic elementary signals used in
  M1.02                                                                  4
Understanding
             communication.
  M1.03      Select various types of signals.                            4
Apply

  M1.04      Explain the basic operation on signals.                     3
Understanding

 Contents:
  ▯ Define signals: state the importance of signals and systems in the field of
science and
    engineering.
  ▯ Basic elementary signals: Unit step function, unit impulse function, ramp,
parabolic,
    signum, exponential, rectangular, triangular, and sinusoidal.
  ▯ Classification of signals: continuous time and discrete time, deterministic
and non-
    deterministic, even and odd, periodic and aperiodic, energy and power, real
and
    imaginary.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Classify and compare continuous time and discrete time systems

This module is presented from the official syllabus scope for Signals and Systems. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Classify and compare continuous time and discrete time systems
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Show time domain representation of a system. 3 Understanding Compare continuous time and discrete time

Show time domain representation of a system. 3 Understanding Compare continuous time and discrete time is an official topic in Module 2 of Signals and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Show time domain representation of a system. 3 Understanding Compare continuous time and discrete time using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, show time domain representation of a system. 3 understanding compare continuous time and discrete time should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Show time domain representation of a system. 3 Understanding Compare continuous time and discrete time means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Show time domain representation of a system. 3 Understanding Compare continuous time and discrete time □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

3 Understanding systems Interpret impulse response of a continuous

3 Understanding systems Interpret impulse response of a continuous is an official topic in Module 2 of Signals and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding systems Interpret impulse response of a continuous using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding systems interpret impulse response of a continuous should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding systems Interpret impulse response of a continuous means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 3 Understanding systems Interpret impulse response of a continuous [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Understanding time and discrete time system.

3 Understanding time and discrete time system. is an official topic in Module 2 of Signals and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding time and discrete time system. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding time and discrete time system. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding time and discrete time system. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 3 Understanding time and discrete time system. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Identify various properties of systems. 3 Apply Representation of systems: Differential equation representation, Difference equation representation Continuous time and discrete time systems – Impulse response, examples Properties of systems – linearity, time invariant system, invertible, casual and non- casual, stable and unstable.

Identify various properties of systems. 3 Apply Representation of systems: Differential equation representation, Difference equation representation Continuous time and discrete time systems – Impulse response, examples Properties of systems – linearity, time invariant system, invertible, casual and non- casual, stable and unstable. is an official topic in Module 2 of Signals and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify various properties of systems. 3 Apply Representation of systems: Differential equation representation, Difference equation representation Continuous time and discrete time systems – Impulse response, examples Properties of systems – linearity, time invariant system, invertible, casual and non- casual, stable and unstable. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify various properties of systems. 3 apply representation of systems: differential equation representation, difference equation representation continuous time and discrete time systems – impulse response, examples properties of systems – linearity, time invariant system, invertible, casual and non- casual, stable and unstable. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Aspect	What to prepare
Definition/identity	What Identify various properties of systems. 3 Apply □ Representation of systems: Differential equation representation, Difference equation representation □ Continuous time and discrete time systems – Impulse response, examples □ Properties of systems – linearity, time invariant system, invertible, casual and non- casual, stable and unstable. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Identify various properties of systems. 3 Apply □ Representation of systems: Differential equation representation, Difference equation representation □ Continuous time and discrete time systems – Impulse response, examples □ Properties of systems – linearity, time invariant system, invertible, casual and non- casual, stable and unstable. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Classify and compare continuous time and discrete time systems

  M2.01      Show time domain representation of a system.            3
Understanding
             Compare continuous time and discrete time
  M2.02                                                              3
Understanding
             systems
            Interpret impulse response of a continuous
 M2.03                                                             3
Understanding
            time and discrete time system.

 M2.04      Identify various properties of systems.                3
Apply

            Series Test – I                                        1

Contents:
  Representation of systems: Differential equation representation, Difference
equation
   representation
  Continuous time and discrete time systems – Impulse response, examples
  Properties of systems – linearity, time invariant system, invertible, casual and
non-
   casual, stable and unstable.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Explain Fourier representation of signals

This module is presented from the official syllabus scope for Signals and Systems. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain Fourier representation of signals
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

5 Apply series

5 Apply series is an official topic in Module 3 of Signals and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Apply series using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 apply series should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Apply series means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 5 Apply series □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

Summarize the properties of Fourier series 3 Understanding Explain sampling theorem, aliasing,

Summarize the properties of Fourier series 3 Understanding Explain sampling theorem, aliasing, is an official topic in Module 3 of Signals and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize the properties of Fourier series 3 Understanding Explain sampling theorem, aliasing, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize the properties of fourier series 3 understanding explain sampling theorem, aliasing, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize the properties of Fourier series 3 Understanding Explain sampling theorem, aliasing, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Summarize the properties of Fourier series 3 Understanding Explain sampling theorem, aliasing, □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

4 Understanding reconstruction Apply Fourier transform, discrete time Fourier

4 Understanding reconstruction Apply Fourier transform, discrete time Fourier is an official topic in Module 3 of Signals and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding reconstruction Apply Fourier transform, discrete time Fourier using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding reconstruction apply fourier transform, discrete time fourier should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding reconstruction Apply Fourier transform, discrete time Fourier means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 4 Understanding reconstruction Apply Fourier transform, discrete time Fourier □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

4 Apply transform Fourier representation of four class of signals Continuous time periodic signal : Fourier series (FS) Discrete time periodic signal : Discrete time Fourier series (DTFS) Continuous time non-periodic signal : Fourier transform (FT) Discrete time non-periodic signal : Discrete time Fourier transform (DTFT) Properties of Fourier representation – linearity, symmetry, time shift, frequency shift, scaling, differentiation and integration, convolution and modulation Sampling theorem, aliasing, reconstruction Apply Laplace transform to demonstrate the concepts of signals and

4 Apply transform Fourier representation of four class of signals Continuous time periodic signal : Fourier series (FS) Discrete time periodic signal : Discrete time Fourier series (DTFS) Continuous time non-periodic signal : Fourier transform (FT) Discrete time non-periodic signal : Discrete time Fourier transform (DTFT) Properties of Fourier representation – linearity, symmetry, time shift, frequency shift, scaling, differentiation and integration, convolution and modulation Sampling theorem, aliasing, reconstruction Apply Laplace transform to demonstrate the concepts of signals and is an official topic in Module 3 of Signals and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Apply transform Fourier representation of four class of signals Continuous time periodic signal : Fourier series (FS) Discrete time periodic signal : Discrete time Fourier series (DTFS) Continuous time non-periodic signal : Fourier transform (FT) Discrete time non-periodic signal : Discrete time Fourier transform (DTFT) Properties of Fourier representation – linearity, symmetry, time shift, frequency shift, scaling, differentiation and integration, convolution and modulation Sampling theorem, aliasing, reconstruction Apply Laplace transform to demonstrate the concepts of signals and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 apply transform □ fourier representation of four class of signals □ continuous time periodic signal : fourier series (fs) □ discrete time periodic signal : discrete time fourier series (dtfs) □ continuous time non-periodic signal : fourier transform (ft) □ discrete time non-periodic signal : discrete time fourier transform (dtft) □ properties of fourier representation – linearity, symmetry, time shift, frequency shift, scaling, differentiation and integration, convolution and modulation □ sampling theorem, aliasing, reconstruction apply laplace transform to demonstrate the concepts of signals and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Apply transform □ Fourier representation of four class of signals □ Continuous time periodic signal : Fourier series (FS) □ Discrete time periodic signal : Discrete time Fourier series (DTFS) □ Continuous time non-periodic signal : Fourier transform (FT) □ Discrete time non-periodic signal : Discrete time Fourier transform (DTFT) □ Properties of Fourier representation – linearity, symmetry, time shift, frequency shift, scaling, differentiation and integration, convolution and modulation □ Sampling theorem, aliasing, reconstruction Apply Laplace transform to demonstrate the concepts of signals and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 4 Apply transform □ Fourier representation of four class of signals □ Continuous time periodic signal : Fourier series (FS) □ Discrete time periodic signal : Discrete time Fourier series (DTFS) □ Continuous time non-periodic signal : Fourier transform (FT) □ Discrete time non-periodic signal : Discrete time Fourier transform (DTFT) □ Properties of Fourier representation – linearity, symmetry, time shift, frequency shift, scaling, differentiation and integration, convolution and modulation □ Sampling theorem, aliasing, reconstruction Apply Laplace transform to demonstrate the concepts of signals and □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Explain Fourier representation of signals
            Apply Fourier series and discrete time Fourier
 M3.01                                                             5
Apply
            series
 M3.02      Summarize the properties of Fourier series             3
Understanding
            Explain sampling theorem, aliasing,
 M3.03                                                             4
Understanding
            reconstruction
            Apply Fourier transform, discrete time Fourier
 M3.04                                                             4
Apply
            transform

Contents:
 □ Fourier representation of four class of signals
    □ Continuous time periodic signal : Fourier series (FS)
    □ Discrete time periodic signal : Discrete time Fourier series (DTFS)
    □ Continuous time non-periodic signal : Fourier transform (FT)
    □ Discrete time non-periodic signal : Discrete time Fourier transform (DTFT)
 □ Properties of Fourier representation – linearity, symmetry, time shift,
frequency shift,
   scaling, differentiation and integration, convolution and modulation
 □ Sampling theorem, aliasing, reconstruction
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

systems

This module is presented from the official syllabus scope for Signals and Systems. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: systems
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

5 Understanding a signal using Laplace transform

5 Understanding a signal using Laplace transform is an official topic in Module 4 of Signals and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding a signal using Laplace transform using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding a signal using laplace transform should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding a signal using Laplace transform means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 5 Understanding a signal using Laplace transform □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Illustrate the region of convergence

Illustrate the region of convergence is an official topic in Module 4 of Signals and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the region of convergence using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the region of convergence should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the region of convergence means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Illustrate the region of convergence [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Outline Properties of Laplace transform

Outline Properties of Laplace transform is an official topic in Module 4 of Signals and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline Properties of Laplace transform using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline properties of laplace transform should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline Properties of Laplace transform means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Outline Properties of Laplace transform [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Apply Inverse Laplace transform to Signals 5 Applying Need of Laplace transform Region of Convergence (ROC) Advantages and limitation of Laplace transform Laplace transform of some commonly used signals - impulse, step, ramp, parabolic, exponential, sine and cosine functions Properties of Laplace transform: Linearity, time shifting, time scaling, time reversal, transform of derivatives and integrals, initial value theorem, final value theorem. Inverse Laplace transform: simple problems (no derivation required) Text/Reference: T/R Book Title/Author T1 Simon Haykin : Signals and System, John Wiley 2/e 2003 T2 A. Anand Kumar : Signals and System, PHI, 2/e 2012 T3 Nagoor Kani : Signals and System, Tata McGrew Hill, 3/e 2011 R1 Ramesh Babu : Signals and Systems, Scotch Publications, 4/e R2 Alan V. Oppenhein, Alan S. Willsky : Signals and System, PHI, 2/e R3 Simon Haykin : Communication Systems, John Wiley 4/e R4 Hwei P. Hsue : Signals and System, Tata McGrew Hill 1995 Rodger E. Ziemer : Signals and System – Continuous and Discrete 4/e, Pearson R5 Education R6 M. J. Robert : Signals and Systems, Tata McGrew Hill, 2003 Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/108/104/108104100/ 2 https://nptel.ac.in/courses/117/101/117101055/

Apply Inverse Laplace transform to Signals 5 Applying Need of Laplace transform Region of Convergence (ROC) Advantages and limitation of Laplace transform Laplace transform of some commonly used signals - impulse, step, ramp, parabolic, exponential, sine and cosine functions Properties of Laplace transform: Linearity, time shifting, time scaling, time reversal, transform of derivatives and integrals, initial value theorem, final value theorem. Inverse Laplace transform: simple problems (no derivation required) Text/Reference: T/R Book Title/Author T1 Simon Haykin : Signals and System, John Wiley 2/e 2003 T2 A. Anand Kumar : Signals and System, PHI, 2/e 2012 T3 Nagoor Kani : Signals and System, Tata McGrew Hill, 3/e 2011 R1 Ramesh Babu : Signals and Systems, Scotch Publications, 4/e R2 Alan V. Oppenhein, Alan S. Willsky : Signals and System, PHI, 2/e R3 Simon Haykin : Communication Systems, John Wiley 4/e R4 Hwei P. Hsue : Signals and System, Tata McGrew Hill 1995 Rodger E. Ziemer : Signals and System – Continuous and Discrete 4/e, Pearson R5 Education R6 M. J. Robert : Signals and Systems, Tata McGrew Hill, 2003 Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/108/104/108104100/ 2 https://nptel.ac.in/courses/117/101/117101055/ is an official topic in Module 4 of Signals and Systems. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Apply Inverse Laplace transform to Signals 5 Applying Need of Laplace transform Region of Convergence (ROC) Advantages and limitation of Laplace transform Laplace transform of some commonly used signals - impulse, step, ramp, parabolic, exponential, sine and cosine functions Properties of Laplace transform: Linearity, time shifting, time scaling, time reversal, transform of derivatives and integrals, initial value theorem, final value theorem. Inverse Laplace transform: simple problems (no derivation required) Text/Reference: T/R Book Title/Author T1 Simon Haykin : Signals and System, John Wiley 2/e 2003 T2 A. Anand Kumar : Signals and System, PHI, 2/e 2012 T3 Nagoor Kani : Signals and System, Tata McGrew Hill, 3/e 2011 R1 Ramesh Babu : Signals and Systems, Scotch Publications, 4/e R2 Alan V. Oppenhein, Alan S. Willsky : Signals and System, PHI, 2/e R3 Simon Haykin : Communication Systems, John Wiley 4/e R4 Hwei P. Hsue : Signals and System, Tata McGrew Hill 1995 Rodger E. Ziemer : Signals and System – Continuous and Discrete 4/e, Pearson R5 Education R6 M. J. Robert : Signals and Systems, Tata McGrew Hill, 2003 Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/108/104/108104100/ 2 https://nptel.ac.in/courses/117/101/117101055/ using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, apply inverse laplace transform to signals 5 applying need of laplace transform region of convergence (roc) advantages and limitation of laplace transform laplace transform of some commonly used signals - impulse, step, ramp, parabolic, exponential, sine and cosine functions properties of laplace transform: linearity, time shifting, time scaling, time reversal, transform of derivatives and integrals, initial value theorem, final value theorem. inverse laplace transform: simple problems (no derivation required) text/reference: t/r book title/author t1 simon haykin : signals and system, john wiley 2/e 2003 t2 a. anand kumar : signals and system, phi, 2/e 2012 t3 nagoor kani : signals and system, tata mcgrew hill, 3/e 2011 r1 ramesh babu : signals and systems, scotch publications, 4/e r2 alan v. oppenhein, alan s. willsky : signals and system, phi, 2/e r3 simon haykin : communication systems, john wiley 4/e r4 hwei p. hsue : signals and system, tata mcgrew hill 1995 rodger e. ziemer : signals and system – continuous and discrete 4/e, pearson r5 education r6 m. j. robert : signals and systems, tata mcgrew hill, 2003 online resources: sl.no website link 1 https://nptel.ac.in/courses/108/104/108104100/ 2 https://nptel.ac.in/courses/117/101/117101055/ should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Apply Inverse Laplace transform to Signals 5 Applying Need of Laplace transform Region of Convergence (ROC) Advantages and limitation of Laplace transform Laplace transform of some commonly used signals - impulse, step, ramp, parabolic, exponential, sine and cosine functions Properties of Laplace transform: Linearity, time shifting, time scaling, time reversal, transform of derivatives and integrals, initial value theorem, final value theorem. Inverse Laplace transform: simple problems (no derivation required) Text/Reference: T/R Book Title/Author T1 Simon Haykin : Signals and System, John Wiley 2/e 2003 T2 A. Anand Kumar : Signals and System, PHI, 2/e 2012 T3 Nagoor Kani : Signals and System, Tata McGrew Hill, 3/e 2011 R1 Ramesh Babu : Signals and Systems, Scotch Publications, 4/e R2 Alan V. Oppenhein, Alan S. Willsky : Signals and System, PHI, 2/e R3 Simon Haykin : Communication Systems, John Wiley 4/e R4 Hwei P. Hsue : Signals and System, Tata McGrew Hill 1995 Rodger E. Ziemer : Signals and System – Continuous and Discrete 4/e, Pearson R5 Education R6 M. J. Robert : Signals and Systems, Tata McGrew Hill, 2003 Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/108/104/108104100/ 2 https://nptel.ac.in/courses/117/101/117101055/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Apply Inverse Laplace transform to Signals 5 Applying Need of Laplace transform Region of Convergence (ROC) Advantages and limitation of Laplace transform Laplace transform of some commonly used signals - impulse, step, ramp, parabolic, exponential, sine and cosine functions Properties of Laplace transform: Linearity, time shifting, time scaling, time reversal, transform of derivatives and integrals, initial value theorem, final value theorem. Inverse Laplace transform: simple problems (no derivation required) Text/Reference: T/R Book Title/Author T1 Simon Haykin : Signals and System, John Wiley 2/e 2003 T2 A. Anand Kumar : Signals and System, PHI, 2/e 2012 T3 Nagoor Kani : Signals and System, Tata McGrew Hill, 3/e 2011 R1 Ramesh Babu : Signals and Systems, Scotch Publications, 4/e R2 Alan V. Oppenhein, Alan S. Willsky : Signals and System, PHI, 2/e R3 Simon Haykin : Communication Systems, John Wiley 4/e R4 Hwei P. Hsue : Signals and System, Tata McGrew Hill 1995 Rodger E. Ziemer : Signals and System – Continuous and Discrete 4/e, Pearson R5 Education R6 M. J. Robert : Signals and Systems, Tata McGrew Hill, 2003 Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/108/104/108104100/ 2 https://nptel.ac.in/courses/117/101/117101055/ □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
systems
            Interpret the frequency domain parameters of
 M4.01                                                            5
Understanding
            a signal using Laplace transform

 M4.02      Illustrate the region of convergence                  2
Understanding

  M4.03     Outline Properties of Laplace transform               4
Understanding
     M4.04    Apply Inverse Laplace transform to Signals            5
Applying

              Series Test – II                                      1

 Contents:
 □     Need of Laplace transform
 □     Region of Convergence (ROC)
 □     Advantages and limitation of Laplace transform
 □     Laplace transform of some commonly used signals - impulse, step, ramp,
parabolic,
       exponential, sine and cosine functions
 □     Properties of Laplace transform: Linearity, time shifting, time scaling,
time reversal,
       transform of derivatives and integrals, initial value theorem, final value
theorem.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Define signals and its importance. 3 Understanding Explain the basic elementary signals used in. (3 marks)

Answer: Begin with the standard definition or identity of *Define signals and its importance. 3 Understanding Explain the basic elementary signals used in*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding communication.. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding communication.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Select various types of signals. 4 Apply. (3 marks)

Answer: Begin with the standard definition or identity of *Select various types of signals. 4 Apply*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain the basic operation on signals. 3 Understanding □ Define signals: state the importance of signals and systems in the field of science and engineering. □ Basic elementary signals: Unit step function, unit impulse function, ramp, parabolic, signum, exponential, rectangular, triangular, and sinusoidal. □ Classification of signals: continuous time and discrete time, deterministic and non- deterministic, even and odd, periodic and aperiodic, energy and power, real and imaginary. □ Mathematical operations on signals: amplitude scaling, time scaling, time shifting, time reversal, addition, subtraction, multiplication and convolution.. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the basic operation on signals. 3 Understanding □ Define signals: state the importance of signals and systems in the field of science and engineering. □ Basic elementary signals: Unit step function, unit impulse function, ramp, parabolic, signum, exponential, rectangular, triangular, and sinusoidal. □ Classification of signals: continuous time and discrete time, deterministic and non- deterministic, even and odd, periodic and aperiodic, energy and power, real and imaginary. □ Mathematical operations on signals: amplitude scaling, time scaling, time shifting, time reversal, addition, subtraction, multiplication and convolution.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain Show time domain representation of a system. 3 Understanding Compare continuous time and discrete time. (3 marks)

Answer: Begin with the standard definition or identity of *Show time domain representation of a system. 3 Understanding Compare continuous time and discrete time*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding systems Interpret impulse response of a continuous. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding systems Interpret impulse response of a continuous*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 3 Understanding time and discrete time system.. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding time and discrete time system.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Identify various properties of systems. 3 Apply □ Representation of systems: Differential equation representation, Difference equation representation □ Continuous time and discrete time systems – Impulse response, examples □ Properties of systems – linearity, time invariant system, invertible, casual and non- casual, stable and unstable.. (3 marks)

Answer: Begin with the standard definition or identity of *Identify various properties of systems. 3 Apply □ Representation of systems: Differential equation representation, Difference equation representation □ Continuous time and discrete time systems – Impulse response, examples □ Properties of systems – linearity, time invariant system, invertible, casual and non- casual, stable and unstable.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Module 3

Define or explain 5 Apply series. (3 marks)

Answer: Begin with the standard definition or identity of *5 Apply series*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Summarize the properties of Fourier series 3 Understanding Explain sampling theorem, aliasing,. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize the properties of Fourier series 3 Understanding Explain sampling theorem, aliasing,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 4 Understanding reconstruction Apply Fourier transform, discrete time Fourier. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding reconstruction Apply Fourier transform, discrete time Fourier*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 4 Apply transform □ Fourier representation of four class of signals □ Continuous time periodic signal : Fourier series (FS) □ Discrete time periodic signal : Discrete time Fourier series (DTFS) □ Continuous time non-

periodic signal : Fourier transform (FT) □ Discrete time non-periodic signal : Discrete time Fourier transform (DTFT) □ Properties of Fourier representation – linearity, symmetry, time shift, frequency shift, scaling, differentiation and integration, convolution and modulation □ Sampling theorem, aliasing, reconstruction Apply Laplace transform to demonstrate the concepts of signals and. (3 marks)

Answer: Begin with the standard definition or identity of *4 Apply transform □ Fourier representation of four class of signals □ Continuous time periodic signal : Fourier series (FS) □ Discrete time periodic signal : Discrete time Fourier series (DTFS) □ Continuous time non-periodic signal : Fourier transform (FT) □ Discrete time non-periodic signal : Discrete time Fourier transform (DTFT) □ Properties of Fourier representation – linearity, symmetry, time shift, frequency shift, scaling, differentiation and integration, convolution and modulation □ Sampling theorem, aliasing, reconstruction Apply Laplace transform to demonstrate the concepts of signals and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

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Module 4

Define or explain 5 Understanding a signal using Laplace transform. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding a signal using Laplace transform*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Illustrate the region of convergence. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate the region of convergence*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Outline Properties of Laplace transform. (3 marks)

Answer: Begin with the standard definition or identity of *Outline Properties of Laplace transform*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Apply Inverse Laplace transform to Signals 5 Applying □ Need of Laplace transform □ Region of Convergence (ROC) □ Advantages and limitation of Laplace transform □ Laplace transform of some commonly used signals - impulse, step, ramp, parabolic, exponential, sine and cosine functions □ Properties of Laplace transform: Linearity, time shifting, time scaling, time reversal, transform of derivatives and integrals, initial value theorem, final value theorem. □ Inverse Laplace transform: simple problems (no derivation required) Text/Reference: T/R Book Title/Author T1 Simon Haykin : Signals and System, John Wiley 2/e 2003 T2 A. Anand Kumar : Signals and System, PHI, 2/e 2012 T3 Nagoor Kani : Signals and System, Tata McGrew Hill, 3/e 2011 R1 Ramesh Babu : Signals and Systems, Scotch Publications, 4/e R2 Alan V. Oppenhein, Alan S. Willsky : Signals and System, PHI, 2/e R3 Simon Haykin : Communication Systems, John Wiley 4/e R4 Hwei P. Hsue : Signals and System, Tata McGrew Hill 1995 Rodger E. Ziemer : Signals and System – Continuous and Discrete 4/e, Pearson R5 Education R6 M. J. Robert : Signals and Systems, Tata McGrew Hill, 2003 Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/108/104/108104100/ 2 https://nptel.ac.in/courses/117/101/117101055/. (3 marks)

Answer: Begin with the standard definition or identity of *Apply Inverse Laplace transform to Signals 5 Applying □ Need of Laplace transform □ Region of Convergence (ROC) □ Advantages and limitation of Laplace transform □ Laplace transform of some commonly used signals - impulse, step, ramp, parabolic, exponential, sine and cosine functions □ Properties of Laplace transform: Linearity, time shifting, time scaling, time reversal, transform of derivatives and integrals, initial value theorem, final value theorem. □ Inverse Laplace transform: simple problems (no derivation required)*

Text/Reference: T/R Book Title/Author T1 Simon Haykin : Signals and System, John Wiley 2/e 2003 T2 A. Anand Kumar : Signals and System, PHI, 2/e 2012 T3 Nagoor Kani : Signals and System, Tata McGrew Hill, 3/e 2011 R1 Ramesh Babu : Signals and Systems, Scotch Publications, 4/e R2 Alan V. Oppenhein, Alan S. Willsky : Signals and System, PHI, 2/e R3 Simon Haykin : Communication Systems, John Wiley 4/e R4 Hwei P. Hsue : Signals and System, Tata McGrew Hill 1995 Rodger E. Ziemer : Signals and System – Continuous and Discrete 4/e, Pearson R5 Education R6 M. J. Robert : Signals and Systems, Tata McGrew Hill, 2003 Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/108/104/108104100/ 2 https://nptel.ac.in/courses/117/101/117101055/. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Define signals and its importance. 3 Understanding Explain the basic elementary signals used in**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Show time domain representation of a system. 3 Understanding Compare continuous time and discrete time**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **5 Apply series**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **5 Understanding a signal using Laplace transform**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- https://nptel.ac.in/courses/108/104/108104100/
- https://nptel.ac.in/courses/117/101/117101055/

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 5201. Verify institutional updates against the current official curriculum.